Roll Number: _____

Thapar Institute of Engineering and Technology, Patiala
Department of Computer Science and Engineering
END Semester Test

B.E. COE/CSE: Semester-VII — Course Name: Compiler Construction (UCS 802)
December 11, 2024, at 2PM — Time: 3 Hours M. Marks: 40
Faculty: Shalini Batra, Karun Verma, Rupali Bhardwaj, Geeta Kasana, Shashank Singh

Note: Attempt any five questions. Attempt parts of one question in sequence. Mention the page no of each attempted question on the front of Answer Sheet. Assume any missing Data.

S. No.	Question **(CLO Mapping and Bloom's Level)**	Marks

Q1 Consider the following grammar:

$$S \rightarrow A$$
$$A \rightarrow BC \mid DBC$$
$$B \rightarrow bB \mid \varepsilon$$
$$C \rightarrow c \mid \varepsilon$$
$$D \rightarrow bc \mid d$$

a) Generate the LR(1) set of items for the grammar. 3

b) Generate the ACTION and GOTO Table. 3

c) Check whether the string '*bcc*' can be parsed by the grammar or not. 2

(CO2, L6)

Q2 Consider an attribute grammar representing binary numerals having $G = \{B, \{B, D, \Lambda\}, \{\cdot, \mathbf{1}, \mathbf{0}\}, N\}$, where N is set of Syntax directed translations as:

$B \rightarrow D_1 \cdot D_2$ $\{N.\boldsymbol{val} = D_1.\boldsymbol{val} + D_2.\boldsymbol{val} * \mathbf{2}^{D_2.len}\}$

$D \rightarrow D_1\, \Lambda$ $\{D.\boldsymbol{val} = \mathbf{2} * D_1.\boldsymbol{val} + \Lambda.\boldsymbol{val}\}\{D.\boldsymbol{len} = D_1.\boldsymbol{len} + 1\}$

$D \rightarrow \Lambda$ $\{D.\boldsymbol{val} = \Lambda.\boldsymbol{val}\}\{D.\boldsymbol{len} = \mathbf{1}\}$

$\Lambda \rightarrow 0$ $\{\Lambda.\boldsymbol{val} = \mathbf{0}\}$

$\Lambda \rightarrow 1$ $\{\Lambda.\boldsymbol{val} = \mathbf{1}\}$

a) In the above grammar G, $\boldsymbol{val}$ is the synthesized attribute of B, D and Λ, whereas $\boldsymbol{len}$ is a synthesized attribute of D. Remove left recursion from the above grammar to make it suitable for top-down parsing. 4

b) Evaluate the value of string $1011 \cdot 1011$ using top-down evaluation. 2

c) Draw the annotated parse tree for the above expression evaluation 2

(CO2, L3)

Q3 **a)** Convert the given expression $(c + a * b) - (a * b + d)$ into 4

i. postfix expression
ii. three address code
iii. syntax tree
iv. DAG

b) Write quadruple, triples, and indirect triples for the following expression: 4

$$a = b * - c + b * - c$$

(CO3, L3)

Q4 **a)** For calculating the income tax, the following formula is used by a concern- 3

$$\textit{if } salary > 500000 \textit{ then}$$
$$Tax = salary \times 0.20$$
$$\textit{else}$$
$$Tax = salary \times 0.10$$
$$\textit{endif}$$

Elaborate on how this formula is converted into a parse tree while passing through lexical and syntax analysis in sequence by giving the input and output at each phase.

S. No.	Question **(CLO Mapping and Bloom's Level)**	Marks
	b) Consider the regular expression $\Phi = b(a\|b\|\epsilon)b^*$. Derive the minimized DFA for Φ using Thomson's rules of NFA construction followed by subset construction.	5
	(CO1, CO2, L2)	
Q5	**a)** What is an activation record? Explain how it is related to run-time storage organization.	3
	b) Explain any two-parameter passing scheme with examples.	2
	c) Given the following calling chain: $main \rightarrow A1 \rightarrow A2 \rightarrow A21 \rightarrow A1$	3

```
main()
{
        --------
        A1()
        {
                ---------
                A2();
        }
        A2 ()
        {
                ---------
                 A21()
                {
                        ---------
                        A1();
                }
                A21();
                ---------
        }
        A1();
}
```

i. Specify the access links (static links) and control links (dynamic links) corresponding to the activation records for each function in the call chain.

ii. Clearly explain how each link is determined based on the calling and lexical structure.

(CO3, L5)

Q6 **a)** Consider the following ANSI C code segment:

```
z = x + 3 + y → f1 + y → f2;
for (i = 0; i < 200; i = i + 2)
{  if(z > i)
    {p = p + x + 3;
     q = q + y → f1;
     }
else
    {
      p = p + y → f2;
      q = q + x + 3;
    }
}
```

Assume that the variable y points to a struct (allocated on the heap) containing two fields $f1$ and $f2$ and the local variables x, y, z, p, q and i are allocated registers.

i. Apply Common Subexpression Elimination to optimize the code. 2

ii. Show how many addition and reference operations are performed on the optimized code. 2

b) Explain the following with examples: 2*2

i. Difference between Constant Propagation and Constant Folding

ii. Induction variable elimination in a loop

(CO4, L3)